CHEM1043 General Chemistry II
Assignment 1 (Topics 1 – 2)

Assignment 1 (15 % of total course mark) is given to you on 18 September 2026 (Fri). Please put your work on A4- or F4-sized white paper (blank or lined) and scan your ***handwritten*** *work into one single pdf file. Name the file in the format of 1043A1_<your student number>, eg. 1043A1_3035000000. Submit the file through* ***Moodle Assignment by 6:00 pm on 9 October 2026 (Fri).***

Please refer to your lecture notes and textbook, and the list of physical constants and the periodic table posted on Moodle for data/constants/equations as necessary.

Note that when calculations involved, you should show the steps by which you arrive at the answers. For example, in question 1a below, you will not just give the mass of the beaker but also show the steps you take to arrive at the answer.

1. A beaker has a radius of 5 cm. (4 marks)

   (a) Calculate the mass (kg) if the beaker exerts 8.5 Torr on the surface.

   (b) The beaker is then filled with hexane (d = 0.655 g/cm^3). Calculate the height (cm) of hexane if it exerts the same pressure as filled with 15 cm of dichloromethane (d = 1.33 g/cm^3).

2. A sulfur (S) sample (2460 g) is burned and converted into SO_2 (g) (see the chemical reaction below). What is the volume (L) of SO_2 (g) collected at 50 °C and 910 mmHg? (3 marks)

$$S_{(s)} + O_{2(g)} = SO_{2(g)}$$

3. A 4.5 L container is filled with N_2 (g) at 600 mmHg and 30 °C. 0.765 g of C_6H_{12} (g) is then injected into the container. (4 marks)

   (a) What is the total pressure in the container?

   (b) What are the partial pressures of N_2 and of C_6H_{12}, respectively?

4.

$$2Na_2O_2\ (s) + 2CO_2\ (g) \rightarrow 2Na_2CO_3\ (s) + O_2\ (g)$$

This reaction is used in submarines and space vehicles to remove expired CO_2 (g) and to generate O_2 (g) for breathing. The rate of CO_2 consumed is 0.35 L/min at 25 °C and 800 mmHg.

If the CO_2 (g) and O_2 (g) in the above reaction are measured at the same temperature and pressure: (6 marks)

(a) What volume (mL) of O_2 (g) is produced per minute? Show your calculations.

(b) What is the rate of Na_2O_2 (s) consumed in grams per hour? Show your calculations.

5. A balloon was filled with 40 mL of N_2 (g) at an initial pressure (P_i). When the pressure is increased by 120 mmHg, the volume becomes 23.1 mL. (5 marks)

   (a) Assuming constant temperature and no leakage, calculate P_i and the final pressure (P_f).
   (b) The sealed balloon is taken to a high altitude and it inflates almost to bursting. Using the kinetic-molecular theory, explain how the density would change and the phenomenon using kinetic-molecular theory of gas.

6. A 5.2 L rigid cylinder contains H_2 (g) at 10.5 atm and 22 °C. Assuming ideal gas behavior and constant temperature, what mass of Ar (g) must be added to raise the total pressure to 90.2 atm? (3 marks)

7. Two ideal gases, A and B, are mixed in a 3 L container at 285 K. Before mixing, A occupies 18.8 L at 2.33 bar and 325 K; B occupies 3.72 L at 50.2 bar and 299 K. Assuming no reaction between A and B, calculate the final partial pressures (bar) of A and B, and the total pressure. Show your working. (5 marks)

8. Using kinetic-molecular theory of gases, explain why a gas fills uniformly a container of any shape. (4 marks)

9. 3.5 kg of dry ice (solid CO_2) is completely sublimed and collected over water at 21 °C and 1 atm. Calculate the partial pressure of CO_2 (in Torr) and the volume of CO_2 collected (in L). (4 marks)

Vapour Pressure of Water at Various Temperatures

	Vapor Pressure	
t,°C	kPa	mmHg
19.0	2.198	16.49
21.0	2.448	18.66
23.0	2.811	21.08
25.0	3.170	23.78

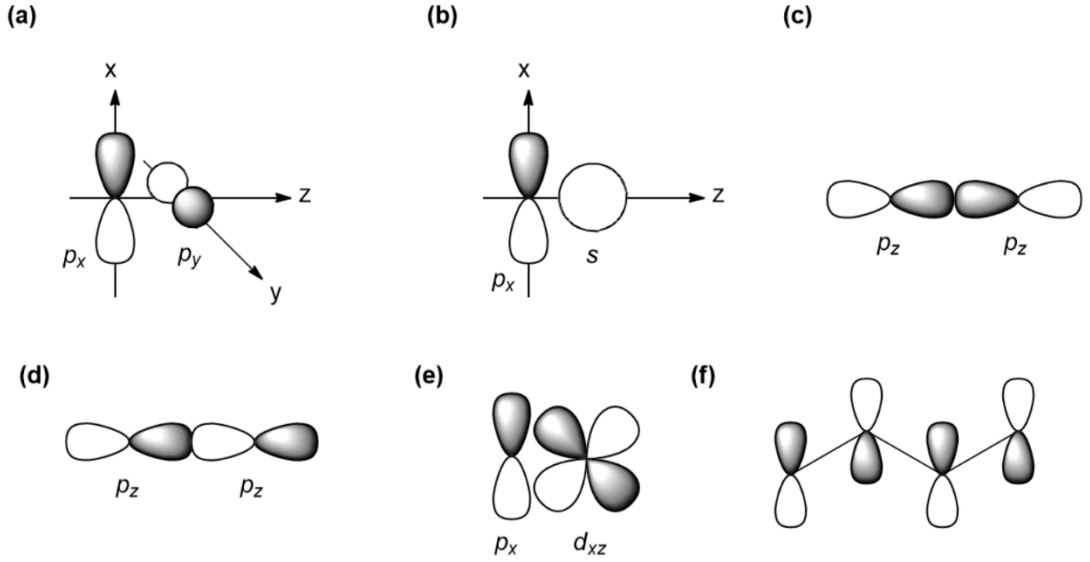
10. Calculate the average kinetic energy, $\overline{E_k}$, for O_2 (g) at 325 K and 1.00 atm. Show your calculations. (3 marks)

11. Consider a sample of H_2S (g) at 25°C and 4.9 atm. Calculate the u_{rms} and use the equation bellow to calculate fraction (F) of molecules that have speed equal to this u_{rms}. (5 marks)

$$F(u) = 4\pi(\frac{M}{2\pi RT})^{\frac{3}{2}} \times u^2 \times e^{-\frac{Mu^2}{2RT}}$$

12. Use both the van der Waals equation and ideal gas equation to calculate the pressure (in bar) exerted by 5 mol CH_4 (g) in a volume of 11.0 L at 298 K. For CH_4, a = 2.303 bar L^2 mol^{-2} and b = 0.0431 L mol^{-1}. (4 marks)

13. For each atomic orbital combination below, assume the orbitals are close enough in energy and overlap sufficiently to interact where symmetry allows. Classify any resulting MO as bonding, non-bonding, or antibonding. For bonding or antibonding MOs, label them as σ or π. (6 marks)



14. Consider CO, O_2^- and B_2. (14 marks)

(a) Represent the bonding with a fully labelled MO diagrams and atomic orbitals (AOs) in each case.

(b) Predict the bond order and magnetic property in each case.

(c) Compare the bond lengths and their relative stability (towards dissociation) of these three species. Explain briefly your answer.

15. Draw the fully labeled π-system MO diagram and shapes of the MOs of benzene (4 marks). Indicate nodes in your drawings (1 mark). Calculate the total bond order for each C-C bond (1 mark).

16. Draw the fully labeled π-system MO diagram and shapes of the MOs of ozone (4 marks). Indicate nodes in your drawings (1 mark). Calculate the total bond order for each O-O bond (1 mark).

17. Draw the fully labeled π-system MO diagram and shapes of the MOs of s-trans-1,3-butadiene (4 marks). Indicate nodes in your drawings (1 mark). Calculate the total π bond order of the π-system (1 marks).

18. Draw the five d-orbitals of Fe^{2+} in hemoglobin. The d_z^2 and d_x^2-$_y^2$ orbitals are degenerate and have higher energy than d_{xy}, d_{yz}, d_{xz} orbitals, which are also degenerate. Draw electron configurations in the d orbitals. Indicate which orbitals can form a sigma bond with carbon monoxide (CO) and can donate π electrons to CO, respectively. Indicate corresponding MOs of CO making the σ and π bond, respectively. (5 marks)

19. Use band theory to explain the electrical conductivity of metals, semi-conductors and insulators, and explain how temperature affects the conductivity of semi-conductors. (7 marks)

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